A regular-modulo-ideal subcase of the trace commutator problem

Candidate partial result for arXiv:1712.06702

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Status. Candidate partial result, likely valid. The unrestricted question remains open. The proof below covers every pair for which either factor is von Neumann regular modulo the trace ideal; in particular it covers every pair with a closed-range factor.

1 Source question

Loreaux and Weiss ask the following as Question 1 of [1]. Here $\mathcal{J}$ is a proper two-sided ideal of $\mathcal{B}(\mathcal{H})$ and a trace is a linear functional $\tau : \mathcal{J} \to \mathbb{C}$ satisfying $\tau(XS) = \tau(SX)$ for $X \in \mathcal{J}$ and $S \in \mathcal{B}(\mathcal{H})$.

> Given a trace $\tau : \mathcal{J} \to \mathbb{C}$ and $A, B \in \mathcal{B}(\mathcal{H})$, does
>
> $$\tau(AB) = \tau(BA) \qquad \text{whenever } AB, BA \in \mathcal{J}?$$

The statement appears on page 1 of the source paper:

selfadjoint operator of norm less than or equal to one is the sum of a unitary operator and its adjoint. This follows readily from the continuous function calculus. Suppose $A = A^*$ with $\|A\| \le 1$. Then let $U := A + i\sqrt{I - A^2}$. It is trivial to check that $U + U^* = A$ and $U^*U = UU^* = I$.

The threefold purpose of this paper is to pose the question: for which traces/ideals do all $B(\mathcal{H})$ operators commute under only the condition that their two products AB, BA both lie in the ideal; to provide evidence in support of it; and to point the reader where to look for possible counterexamples.

Question 1. Given a trace $\tau : \mathcal{J} \to \mathbb{C}$ and $A, B \in B(\mathcal{H})$ does

$$\tau(AB) = \tau(BA) \quad \text{whenever } AB, BA \in \mathcal{J}?$$

We now describe our motivation for this question. Of course, the product of two Hilbert–Schmidt operators is trace-class (by the Cauchy–Schwarz inequality), and it is a well-known fact that whenever A, B are Hilbert–Schmidt, then $\operatorname{tr}(AB) = \operatorname{tr}(BA)$; a direct application of Fubini's theorem. Perhaps less well-known is that this extends to the case when $A \in \mathcal{K}$ and $B \in B(\mathcal{H})$ and both products are trace-class. This appears, for instance, in Gohberg–Krein [GK69, Theorem III.8.2] using the Schmidt decomposition for compact operators, or in Dunford–Schwartz [DS63, Lemma XI.9.14(b)]. However, we can obtain an affirmative answer to the stronger Question 1 for the standard trace using a theorem of Lidskii [Lid65] (see Theorem 2) and Theorem 5. Then,

The source proves an affirmative answer for spectral traces and points to nonspectral traces and quasinilpotent operators as possible sources of a counterexample. The theorem below gives a different, purely algebraic affirmative subcase.

2 Idea of the proof

The key operation is to replace one factor A by its defect from having an inner inverse. For an arbitrary bounded C, set $R = A - ACA$. Although neither A nor B need lie in the ideal, a pair of legal cyclic rotations shows that the trace defect is preserved exactly:

$$\tau(AB) - \tau(BA) = \tau(RB) - \tau(BR).$$

If the coset of A is von Neumann regular modulo $\mathcal{J}$, then C can be chosen with $R \in \mathcal{J}$, and the right side vanishes by the defining trace property. For a closed-range operator one may take a bounded generalized inverse and obtain $R = 0$.

Two reductions identify what is left. Matrix amplification turns any counterexample into one in which both factors are square-zero. Conversely, after polar reduction to $A \geq 0$, truncated generalized inverses replace A by an arbitrarily small spectral tail without changing the trace defect. Thus the unresolved mechanism is genuinely tied to nonclosed range and to the lack of continuity of an arbitrary trace.

3 The descent identity and the partial theorem

We first state the argument in a unital algebra, since no Hilbert-space geometry is used.

Lemma 1 (Defect descent)**.** *Let $\mathcal{R}$ be a unital complex algebra, let $\mathcal{I} \lhd \mathcal{R}$ be a two-sided ideal, and let $\tau : \mathcal{I} \to \mathbb{C}$ be a trace. If $a, b \in \mathcal{R}$ satisfy $ab, ba \in \mathcal{I}$, then for every $c \in \mathcal{R}$, with $r = a - aca$, all of rb, br lie in $\mathcal{I}$ and*

$$\tau(ab) - \tau(ba) = \tau(rb) - \tau(br). \tag{1}$$

Proof. The ideal property gives

$$rb = ab - acab \in \mathcal{I}, \qquad br = ba - baca \in \mathcal{I}.$$

Moreover $cab \in \mathcal{I}$ and $ba \in \mathcal{I}$. Two applications of the trace identity, each with one factor in $\mathcal{I}$, give

$$\tau(acab) = \tau(caba) = \tau(baca).$$

Expanding $rb - br$ and cancelling these equal middle terms proves (1). □

Recall that $a + \mathcal{I}$ is *von Neumann regular* in $\mathcal{R}/\mathcal{I}$ if there is $c \in \mathcal{R}$ such that

$$a - aca \in \mathcal{I}.$$

Theorem 2 (Regular-modulo-ideal trace commutation)**.** *Under the hypotheses of the lemma, if either $a + \mathcal{I}$ or $b + \mathcal{I}$ is von Neumann regular in $\mathcal{R}/\mathcal{I}$, then*

$$\tau(ab) = \tau(ba).$$

Proof. Suppose $a - aca = r \in \mathcal{I}$. Since $r \in \mathcal{I}$ and $b \in \mathcal{R}$, $\tau(rb) = \tau(br)$. Equation (1) now gives the conclusion. The case in which $b + \mathcal{I}$ is regular follows after interchanging a and b. □

Corollary 3 (Closed-range factor)**.** *Let $\mathcal{J}$ be a proper $\mathcal{B}(\mathcal{H})$-ideal and let τ be any trace on $\mathcal{J}$. Suppose $A, B \in \mathcal{B}(\mathcal{H})$ and $AB, BA \in \mathcal{J}$. If either A or B has closed range, then*

$$\tau(AB) = \tau(BA).$$

In particular this holds if either factor is Fredholm, semi-Fredholm, a projection, a partial isometry, or bounded below on the orthogonal complement of its kernel.

Proof. If A has closed range, its Moore–Penrose inverse C is bounded and $ACA = A$. Thus A is von Neumann regular already in $\mathcal{B}(\mathcal{H})$, and Theorem 2 applies. The argument is symmetric in A, B. □

4 Two exact reductions of the remaining problem

Proposition 4 (Square-zero reduction)**.** *The universal Loreaux–Weiss question is equivalent to its restriction to pairs P, Q satisfying $P^2 = Q^2 = 0$.*

Proof. Only one direction needs proof. Suppose $A, B, \mathcal{J}, \tau$ form a counterexample. On $\mathcal{H} \oplus \mathcal{H}$ define

$$P = \begin{pmatrix} 0 & A \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 0 \\ B & 0 \end{pmatrix}.$$

Then $P^2 = Q^2 = 0$ and

$$PQ = \begin{pmatrix} AB & 0 \\ 0 & 0 \end{pmatrix}, \qquad QP = \begin{pmatrix} 0 & 0 \\ 0 & BA \end{pmatrix} \in M_2(\mathcal{J}).$$

The functional

$$\tau_2((X_{ij})) = \tau(X_{11}) + \tau(X_{22})$$

is a trace on $M_2(\mathcal{J})$. Therefore $\tau_2(PQ) = \tau(AB) \neq \tau(BA) = \tau_2(QP)$. Since $\mathcal{H} \oplus \mathcal{H}$ is again a separable infinite-dimensional Hilbert space, this is a counterexample of the required square-zero form. $\square$

Combining Proposition 4 with Corollary 3 shows that a negative answer, if one exists, already exists among square-zero factors whose ranges are both nonclosed.

Proposition 5 (Positive spectral-tail localization)**.** *Suppose $A \geq 0$, $B \in \mathcal{B}(\mathcal{H})$, and $AB, BA \in \mathcal{J}$. For every $\epsilon > 0$ let*

$$E_\epsilon = \mathbf{1}_{[0,\epsilon)}(A), \qquad R_\epsilon = AE_\epsilon.$$

Then $R_\epsilon B, BR_\epsilon \in \mathcal{J}$, $\|R_\epsilon\| \leq \epsilon$, and

$$\Delta_\tau(A, B) := \tau(AB) - \tau(BA) = \tau(R_\epsilon B) - \tau(BR_\epsilon).$$

Moreover, every putative counterexample can first be reduced to one with a positive first factor.

Proof. Define the bounded Borel function

$$f_\epsilon(t) = \begin{cases} t^{-1}, & t \geq \epsilon, \\ 0, & 0 \leq t < \epsilon, \end{cases} \qquad C_\epsilon = f_\epsilon(A).$$

Functional calculus gives $A - AC_\epsilon A = A\mathbf{1}_{[0,\epsilon)}(A) = R_\epsilon$. The descent lemma gives the trace identity and ideal membership, and the norm bound is immediate.

For the final assertion write the polar decomposition $A = U|A|$. Since $|A|B = U^*AB \in \mathcal{J}$, cyclicity gives

$$\tau(AB) = \tau(U|A|B) = \tau(|A|BU).$$

Also $|A|BU = U^*ABU \in \mathcal{J}$ and $BU|A| = BA \in \mathcal{J}$. Hence the trace defect of (A, B) equals that of the pair $(|A|, BU)$, whose first factor is positive. $\square$

5 Verification, limitations, and novelty status

The proof uses only finite algebraic expansions and legal applications of $\tau(XS) = \tau(SX)$ with $X \in \mathcal{J}$. In particular, it assumes no positivity, normality, or continuity of the trace. The two identities a verifier should check first are

$$\tau(acab) = \tau(caba) = \tau(baca)$$

and the diagonal-block values of PQ and QP in Proposition 4. Both have been checked directly.

This is a partial result only. Theorem 2 says nothing when both cosets are nonregular in $\mathcal{B}(\mathcal{H})/\mathcal{J}$. Proposition 5 does not permit a limit as $\epsilon \downarrow 0$, because an arbitrary trace on an algebraic operator ideal may be completely discontinuous. The nonspectral quasinilpotent construction of Dykema–Kalton [2], noted by the source authors as a possible counterexample route, still lacks a factorization whose two products remain in the same ideal.

A bounded search on 11 August 2026 used the exact title and question, the authors' later papers, the phrases "AB, BA in an ideal", "trace commutator property", "closed range", and "von Neumann regular modulo an ideal", and the related arXiv papers 2103.09215 and 2203.11995. It found no resolution of Question 1 and no occurrence of the regular-modulo-ideal theorem above. This is not a claim of definitive novelty; novelty confidence is moderate and a specialist literature review is recommended.

Human-review recommendation. Verify the two cyclic rotations in the descent lemma and the amplification trace in Proposition 4. If accepted, retain this as a substantial partial theorem and structural reduction, not as a full solution.

References

[1] J. Loreaux and G. Weiss, *Traces on ideals and the commutator property*, arXiv:1712.06702, 2017.

[2] K. Dykema and N. J. Kalton, *Spectral characterization of sums of commutators. II*, J. Reine Angew. Math. **504** (1998), 127–137; arXiv:math/9709208.